Probability Limit Theorems

MATH-SHU 350

Yao Xiao

Fall 2023

Contents

1 Measure Theory and Probability Spaces **1**
1.1 Measure Theory . . . 1
1.1.1 Basic Concepts . . . 1
1.1.2 Construction and Characterization of Measures . . . 3
1.1.3 Elementary Inequalities . . . 4
1.1.4 Monotone Convergence Properties . . . 5
1.2 Probability Spaces . . . 6
1.2.1 Independence . . . 8
1.2.2 The Fatou and Borel-Cantelli Lemmas . . . 9

8/28 Lecture

1 Measure Theory and Probability Spaces

1.1 Measure Theory

1.1.1 Basic Concepts

Definition 1.1. Let E be a set, and let $\mathscr{A}$ be a set of subsets of E. We say that

$$\text{if} \quad \varnothing \in \mathscr{A}, \quad A^C \in \mathscr{A},\ A \cup B \in \mathscr{A}, \quad \forall A, B \in \mathscr{A}, \quad \text{then } \mathscr{A} \text{ is an } \textbf{algebra}, \tag{1}$$

$$\text{if} \quad \varnothing \in \mathscr{A}, \quad A \cap B \in \mathscr{A}, \quad \forall A, B \in \mathscr{A}, \quad \text{then } \mathscr{A} \text{ is a } \pi\textbf{-system}, \tag{2}$$

$$\text{if} \quad \varnothing \in \mathscr{A}, \quad A^C \in \mathscr{A},\ \bigcup_{n\in\mathbb{N}} A_n \in \mathscr{A}, \quad \forall A, A_n \in \mathscr{A}, \quad \text{then } \mathscr{A} \text{ is a } \sigma\textbf{-algebra}. \tag{3}$$

Definition 1.2. A pair $(E, \mathscr{E})$, where E is a set and $\mathscr{E}$ is a σ-algebra on E, is called a **measurable space**. In this case, each $A \in \mathscr{E}$ is called an $\mathscr{E}$**-measurable set**.

Definition 1.3. The intersection of all σ-algebras containing $\mathscr{C}$ (there is at least one) is called the σ**-algebra generated by** $\mathscr{C}$, denoted by $\sigma(\mathscr{C})$.

Remark 1.4. Note that $\mathscr{P}(E)$ (the set of all subsets of E) is a σ-algebra, and the intersection of any collection of σ-algebras is a σ-algebra. $\sigma(\mathscr{C})$ is the smallest σ-algebra containing $\mathscr{C}$.

Remark 1.5. It is commonly impossible to write down the typical element of a σ-algebra $\mathscr{E}$, so we try to find π-systems with simpler elements that generate our σ-algebra and work with them instead.

Definition 1.6. A **topological space** E with **topology** $\mathscr{T}$ is a space endowed with a set of open subsets. The **Borel σ-algebra** is then defined as $\mathscr{B}(E) = \sigma(\mathscr{T})$, which is the σ-algebra generated by the family of all open subsets of E. We use the standard abbreviation $\mathscr{B} := \mathscr{B}(\mathbb{R})$.

Example 1.7. Let $\pi(\mathbb{R})$ be the π-system $\pi(\mathbb{R}) := \{(-\infty, x];\ x \in \mathbb{R}\}$. CHECK THAT $\sigma(\pi(\mathbb{R})) = \mathscr{B}(\mathbb{R})$, *i.e.*, $\sigma(\pi(\mathbb{R}))$ is the Borel σ-algebra of $\mathbb{R}$.

Definition 1.8. Let E be a set and $\mathscr{A}$ be an algebra on E. A **set function** is any function $\mu : \mathscr{A} \to [0, \infty]$ with $\mu(\varnothing) = 0$.

Definition 1.9. Let E be a set, $\mathscr{A}$ be an algebra on E, and μ be a set function on $\mathscr{A}$. We say that

$$\mu \text{ is } \textbf{increasing} \quad \text{if} \quad \mu(A) \leq \mu(B), \qquad \forall A, B \in \mathscr{A} \text{ with } A \subseteq B, \tag{4}$$

$$\mu \text{ is } \textbf{additive} \quad \text{if} \quad \mu(A \cup B) = \mu(A) + \mu(B), \qquad \forall A, B \in \mathscr{A} \text{ disjoint}, \tag{5}$$

$$\mu \text{ is } \textbf{σ-additive} \quad \text{if} \quad \mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} \mu(A_n), \qquad \forall \{A_n\}_{n\in\mathbb{N}} \subseteq \mathscr{A} \text{ disjoint, with } \bigcup_{n\in\mathbb{N}} A_n \in \mathscr{A}. \tag{6}$$

Remark 1.10. σ-additivity is also called countable additivity. Let $(E, \mathscr{E})$ be a measurable space, then if $\mu : \mathscr{E} \to [0, \infty]$ is countable additive, it is called a **measure** and the triple $(E, \mathscr{E}, \mu)$ is called a **measure space**. In particular if $\mu(E) = 1$, then μ is called a **probability measure** and $(E, \mathscr{E}, \mu)$ is called a **probability space**. The notation $(\Omega, \mathscr{F}, \mathbb{P})$ is often used instead for probability spaces.

Remark 1.11. Given a measure space $(E, \mathscr{E}, \mu)$, μ is said to be **finite** if $\mu(E) < \infty$, and **σ-finite** if there exists $\{E_n\}_{n\in\mathbb{N}} \subseteq \mathscr{E}$, such that $\mu(E_n) < \infty$ and $\bigcup_{n\in\mathbb{N}} E_n = E$. A set $F \in \mathscr{E}$ is called **μ-null** if $\mu(F) = 0$.

Remark 1.12. A statement ϕ about points $s \in E$ holds **almost everywhere** (*a.e.*) if $F = \{s \in E;\ \phi(s) \text{ is false}\} \in \mathscr{E}$ and F is μ-null, *i.e.*, $\mu(F) = 0$. For a probability space, almost everywhere is often called **almost surely** (*a.s.*).

Example 1.13 (Infinitely often)**.** A_n occurs infinitely often if $\forall p \in \mathbb{N}$, $\exists n \geq p$, such that A_n holds. Now look at the event that an element $\omega \in$ infinitely many A_n's. This is equivalent to $\forall p \in \mathbb{N}$, $\exists n \geq p$, such that $\omega \in A_n$. Then $\forall p \in \mathbb{N}$, $\omega \in \bigcup_{n\geq p} A_n$, *i.e.*, $\omega \in \bigcap_{p\in\mathbb{N}} \bigcup_{n\geq p} A_n$. Therefore, we can see that

$$\{\omega;\ \omega \in A_n \text{ infinitely often}\} = \bigcap_{p\in\mathbb{N}} \bigcup_{n\geq p} A_n. \tag{7}$$

Note that we can interpret unions and intersections as

$$\bigcup_{n\in\mathbb{N}} A_n = \{A_n \text{ occurs at least once}\}, \qquad \bigcap_{n\in\mathbb{N}} A_n = \{\text{all } A_n\text{'s occur}\}. \tag{8}$$

Example 1.14 (A set we cannot measure on $\mathbb{R}$)**.** Take the equivalence class $x \sim y \iff y - x \in \mathbb{Q}$. The axiom of choice enables us to define a set C, such that C as exactly one element in each equivalent class, and we assume that $C \subseteq [0, 1]$ (applicable by translation). Assume $\mu(C) = 0$ and we define for some $r \in \mathbb{R}$ that

$$C_r = C + r = \{x \in \mathbb{R};\ x \in \mathbb{R},\ a \in C\}. \tag{9}$$

We can CHECK THAT C_r, $r \in \mathbb{Q}$ are disjoint. Indeed, if we take $x \in C_r \cap C_s$, then we can write $x = c + r = c' + s$ for some $c, c' \in C$. This would imply that $c' = c + r - s$, meaning that $r = s$ since there is can be only one element in each equivalent class. This leads to a contradiction, so that $C_r \cap C_s = \varnothing$, $\forall r, s \in \mathbb{Q}$. Then we can see that

$$\mu(\mathbb{R}) = \mu\left(\bigcup_{r\in\mathbb{Q}} C_r\right) = \sum_{r\in\mathbb{Q}} \mu(C_r) = \underbrace{\sum_{r\in\mathbb{Q}} \mu(C)}_{\text{translation invariance}} = \sum_{r\in\mathbb{Q}} 0 = 0, \tag{10}$$

which leads to a contradiction. Therefore C is not measurable on $\mathbb{R}$. We shall also be able to CHECK THAT if $\mu(C) > 0$, then $\mu((0, 2)) = \infty$, and in general $\mu((a, b)) = \infty$, $\forall a < b$. Indeed, we have that

$$\mu((0, 2)) \geq \mu\left(\bigcup_{r\in\mathbb{Q}\cap[0,1]} C_r\right) = \sum_{r\in\mathbb{Q}\cap[0,1]} \mu(C_r) = \sum_{r\in\mathbb{Q}\cap[0,1]} \mu(C) = \infty. \tag{11}$$

8/30 Lecture

Example 1.15 (An algebra that is not a σ-algebra)**.** Let $\mathbb{N}$ be the space and take

$$\mathscr{E} = \left\{A \subseteq \mathbb{N};\ A \text{ is finite or } A^C \text{ is finite}\right\}. \tag{12}$$

If $A \in \mathscr{E}$, then either A is finite (which means $(A^C)^C$ is finite) or A^C is finite, implying that $A^C \in \mathscr{E}$. Now if $A, B \in \mathscr{E}$, we discuss two cases. If A and B are both finite, then $A \cup B$ is finite. Otherwise A^C is finite, thus $(A \cup B)^C = A^C \cap B^C$ is finite. We have shown that $\mathscr{E}$ is indeed an algebra. However, if we take $A_n = \{2n\}$, then $A = \bigcup_{n\in\mathbb{N}} A_n$ is the set of all even numbers, which is not finite and does not have a finite complement either. It follows that $\mathscr{E}$ is not a σ-algebra.

Example 1.16. Given $A_1, \cdots, A_m$ subsets of a non-empty set E, and let $\mathscr{A} := \{A_1, \cdots, A_m\}$. The goal is to describe all elements of $\mathscr{E} = \sigma(\mathscr{A})$. To do this, we denote $A_i^1 := A_i$ and $A_i^0 := A_i^C$, then for some $\epsilon \in \{0,1\}^m$, define $A_\epsilon = \bigcap_{1\le i\le m} A_i^{\epsilon_i}$. We shall first show that

$$A \in \mathscr{E} \iff A = \bigcup_{\epsilon\in I} A_\epsilon \quad \text{for some } I \subseteq \{0,1\}^m. \tag{13}$$

$\Longleftarrow$ This is trivial by definition. Each $A_i \in \mathscr{E}$, so each $A_i^C \in \mathscr{E}$, and thus each A_ϵ, being a finite union of A_i's and their complements, is in $\mathscr{E}$. This implies that any countable union of A_ϵ's is also in $\mathscr{E}$.

$\Longrightarrow$ Let

$$\mathscr{F} = \left\{A \subseteq E;\ A = \bigcup_{\epsilon\in I} A_\epsilon \text{ for some } I \subseteq \{0,1\}^m\right\}. \tag{14}$$

Clearly $\varnothing \in \mathscr{F}$. Take an arbitrary $A = \bigcup_{\epsilon\in I} A_\epsilon \in \mathscr{F}$, then we have that

$$A^C = \bigcup_{\epsilon\in I^C} A_\epsilon \in \mathscr{F}. \tag{15}$$

This is because all A_ϵ's are disjoint, and their union is the entire space E. Now take $A_n \in \mathscr{F}$, $n \in \mathbb{N}$, then

$$\bigcup_{n\in\mathbb{N}} A_n = \bigcup_{\epsilon\in\bigcup_{n\in\mathbb{N}} I_n} A_\epsilon \in \mathscr{F}, \tag{16}$$

thus we can conclude that $\mathscr{F}$ is a σ-algebra. Note that $\mathscr{A} \subseteq \mathscr{F}$. Indeed, if we visualize on a Venn diagram, we will see that each A_ϵ represents a single and unique area in that diagram. Each set A_i would be some union of these single areas, so $\mathscr{A} \subseteq \mathscr{F}$. Moreover, $\mathscr{E} = \sigma(\mathscr{A})$ is the smallest σ-algebra containing $\mathscr{A}$, and since $\mathscr{F}$ is an σ-algebra containing $\mathscr{A}$, we can conclude that $\mathscr{E} \subseteq \mathscr{F}$, and the proof is complete.

Now we can see that each element of $\mathscr{E}$ can be written in the form of $A = \bigcup_{\epsilon\in I} A_\epsilon$ for some $I \subseteq \{0,1\}^m$. Note that for all $\epsilon \in \{0,1\}^m$ and $A \in \mathscr{E}$, either $A \cap A_\epsilon = \varnothing$ or $A_\epsilon \subseteq A$. This is the same meaning as "a single area in the Venn diagram" as in the proof above. The subsets A_ϵ are thus called the atoms of $\mathscr{E}$.

1.1.2 Construction and Characterization of Measures

As previously mentioned, it is difficult to directly work with a σ-algebra (for instance, defining a measure). Therefore, we try to specify the values on a π-system generating it, and extend the definition to the whole σ-algebra.

Lemma 1.17. Let E be a set, $\mathscr{I}$ be a π-system on E, and $\mathscr{E} = \sigma(\mathscr{I})$. Suppose that μ_1 and μ_2 are measures on $(E, \mathscr{E})$, such that $\mu_1|_{\mathscr{I}} = \mu_2|_{\mathscr{I}}$ and $\mu_1(E) = \mu_2(E) < \infty$, then $\mu_1 = \mu_2$ on $\mathscr{E}$.

Remark 1.18. Note that if $\mathscr{I}$ is not a π-system, the uniqueness (as specified in the lemma above) can fail. For instance, let $E = \{1,2,3,4\}$ and $\mathscr{I} = \{\{1,2\},\{1,3\}\}$. Let $r_i = \mu_1(\{i\})$ and $s_i = \mu_2(\{i\})$. Assuming that $\mu_1|_{\mathscr{I}} = \mu_2|_{\mathscr{I}}$, we have that $r_1 + r_2 = s_1 + s_2$ and $r_1 + r_3 = s_1 + s_3$. However, this does not necessarily mean that $r_1 + r_2 + r_3 + r_4 = s_1 + s_2 + s_3 + s_4$, so $\mu_1 = \mu_2$ does not necessarily hold on $\mathscr{E}$ (note that $\mathscr{E}$ necessarily contains E because it is a σ-algebra).

Theorem 1.19 (Carathéodory's extension theorem)**.** Let E be a set, $\mathscr{E}_0$ be an algebra on E, and $\mathscr{E} := \sigma(\mathscr{E}_0)$. Assume that $\mu_0 : \mathscr{E}_0 \to [0, \infty]$ is a countably additive set function on $\mathscr{E}_0$, then there exists a measure μ on $(E, \mathscr{E})$, such that $\mu = \mu_0$ on $\mathscr{E}_0$.

Remark 1.20. Lemma 1.17 and Theorem 1.19 are important tools for proving uniqueness and existence of measures (respectively), for instance, the Lebesgue measure.

Theorem 1.21. There exists a unique Borel measure μ on $\mathbb{R}$, such that $\forall a < b \in \mathbb{R}$, $\mu((a, b]) = b - a$. This measure μ is called the **Lebesgue measure on** $\mathbb{R}$.

1.1.3 Elementary Inequalities

Lemma 1.22. Let $(E, \mathscr{E}, \mu)$ be a measure space. Then,

$$\mu(A \cup B) \leq \mu(A) + \mu(B), \qquad \mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B). \tag{17}$$

Proof. We have that

$$\mu(A \cup B) = \mu(A \cup (B \setminus A)) = \mu(A) + \mu(B \setminus A) \leq \mu(A) + \mu(B), \tag{18}$$
$$\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B \setminus A) + \mu(A \cap B) = \mu(A) + \mu(B), \tag{19}$$

so the proof is complete. The second equation in fact implies the first inequality. □

Remark 1.23. Both observations above can be generalized to a finite number of sets. The first inequality follows by the observation that $\bigcup_{n \in \mathbb{N}} F_n = \bigcup_{n \in \mathbb{N}} \left(F_n \setminus \bigcup_{i \leq n-1} F_i \right)$. The second equality follows by induction into the inclusion-exclusion formula.

9/1 Recitation

Example 1.24. Let Ω be a non-empty set. If $\mathscr{F}_i$ is a σ-algebra for each $i \in I$, where I is a non-empty set of indices. Show that $\mathscr{F} = \bigcap_{i \in I} \mathscr{F}_i$ is a σ-algebra.

Proof. We have that $\varnothing, \Omega \in \mathscr{F}_i$, $\forall i \in I$, thus $\varnothing, \Omega \in \mathscr{F}$. Take an arbitrary $A \in \mathscr{F}$, we have that $A \in \mathscr{F}_i$, $\forall i \in I$, and thus $A^C \in \mathscr{F}_i$, $\forall i \in I$, so that $A^C \in \mathscr{F}$. Now take arbitrary $\{A_n\}_{n \geq 1} \subseteq \mathscr{F}$, then $A_n \in \mathscr{F}_i$, $\forall i \in I$, $\forall n \geq 1$. Therefore, $\bigcup_{n \geq 1} A_n \in \mathscr{F}_i$, $\forall i \in I$. This means that $\bigcup_{n \geq 1} A_n \in \mathscr{F}$, and the proof is complete. □

Example 1.25. Let Ω be a non-empty set.

(1) Show that if $\mathscr{F}_1 \subset \mathscr{F}_2 \subset \cdots$ are σ-algebras, then $\mathscr{F} = \bigcup_{n \geq 1} \mathscr{F}_n$ is an algebra.

(2) Show that $\mathscr{F}$ is not a σ-algebra.

Proof. (1) We have that $\varnothing, \Omega \in \mathscr{F}_n$, $\forall n$. Therefore, $\varnothing, \Omega \in \mathscr{F}$. Take an arbitrary $A \in \bigcup_{n \geq 1} \mathscr{F}_n$, then there exists $n_0 \geq 1$, such that $A \in \mathscr{F}_{n_0}$. This means that $A^C \in \mathscr{F}_{n_0}$, and thus $A^C \in \mathscr{F}$. Now take arbitrary $A, B \in \mathscr{F}$, then there exists $n_1, n_2 \geq 1$, such that $A \in \mathscr{F}_{n_1}$ and $B \in \mathscr{F}_{n_2}$. We have that $A \cup B \in \mathscr{F}_{\max(n_1, n_2)}$, so that $A \cup B \in \mathscr{F}$. This completes the proof.

(2) Take a counterexample $\mathscr{F}_n = \mathscr{P}(\{1, \cdots, n\})$, the set of all subsets of $\{1, \cdots, n\}$. We can see that $\mathscr{F}$ is then the set of subsets of $\mathbb{N}$ with *finite* cardinality. Now consider $A_n = \{n\} \in \mathscr{F}_n$, $n \geq 1$, then the countable union $\bigcup_{n \geq 1} A_n = \mathbb{N} \notin \mathscr{F}$. Therefore $\mathscr{F}$ is not a σ-algebra in this case. □

Example 1.26. Let μ be a finitely additive measure in an algebra $\mathscr{S}$. Consider $(A_n)_{n \geq 1}$ disjoint sets in $\mathscr{S}$ such that $A = \bigcup_{n \geq 1} A_n \in \mathscr{S}$. Show that

$$\mu(A) \geq \sum_{n=1}^{\infty} \mu(A_n). \tag{20}$$

Proof. Note that $\mathscr{S}$ is not necessarily a σ-algebra. Decompose

$$A = \bigcup_{n=1}^{\infty} A_n = \underbrace{\left(\bigcup_{n=1}^{k} A_n\right)}_{\in \mathscr{S}} \cup \underbrace{\left(\bigcup_{n=k+1}^{\infty} A_n\right)}_{\in \mathscr{S}}. \tag{21}$$

Therefore, we can see that

$$\mu(A) = \mu\left(\bigcup_{n=1}^{k} A_n\right) + \underbrace{\mu\left(\bigcup_{n=k+1}^{\infty} A_n\right)}_{\geq 0} \geq \mu\left(\bigcup_{n=1}^{k} A_n\right). \tag{22}$$

If $\mu(A) = \infty$, we are done, so we assume that $\mu(A) < \infty$. Taking $k \to \infty$, we can conclude that

$$\mu(A) \geq \mu\left(\bigcup_{n=1}^{\infty} A_n\right). \tag{23}$$

□

Example 1.27. Consider a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and a family of measurable sets $(A_n)_{n\geq 1}$ such that $\mathbb{P}(A_n) = 1$. Show that

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = 1. \tag{24}$$

Proof. We can see that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n^C\right) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n^C) = \sum_{n=1}^{\infty} 0 = 0, \tag{25}$$

so the proof is complete by taking the complement. □

Example 1.28. For a sequence of events $(A_n)_{n\geq 1}$ with $\lim_{n\to\infty} \mathbb{P}(A_n) = 1$ and $0 < c < 1$, show that there exists a subsequence $(n_k)_{k\geq 1}$ with $n_k \to \infty$, such that

$$\mathbb{P}\left(\bigcap_{k=1}^{\infty} A_{n_k}\right) > c. \tag{26}$$

Proof. Fix $\epsilon_0 = 1 - c > 0$, and take $\epsilon_i = \epsilon_0/2^i$, $\forall i \geq 1$. From the given limit, we can see that $\forall \epsilon > 0$, $\exists n_0 \geq 1$, such that $\forall n \geq n_0$, we have that $|1 - \mathbb{P}(A_n)| < \epsilon$. Therefore, we can take n_k such that $|1 - \mathbb{P}(A_{n_k})| < \epsilon_k$. Then, we have that

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_{n_k}^C\right) \leq \sum_{k=1}^{\infty} \mathbb{P}(A_{n_k}^C) < \sum_{k=1}^{\infty} \frac{\epsilon_0}{2^k} = \epsilon_0 = 1 - c. \tag{27}$$

Then we can conclude that

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_{n_k}\right) = 1 - \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_{n_k}^C\right) > c \tag{28}$$

so the proof is complete. □

9/4 Lecture

1.1.4 Monotone Convergence Properties

Theorem 1.29. Let $(E, \mathscr{E}, \mu)$ be a measure space.

(1) If $\{F_n\}_{n\in\mathbb{N}} \subseteq \mathscr{E}$ and $F_n \uparrow F$, then $\mu(F_n) \uparrow \mu(F)$.

(2) If $\{G_n\}_{n\in\mathbb{N}} \subseteq \mathscr{E}$ and $G_n \downarrow G$ with $\mu(G_k) < \infty$ for some $k \in \mathbb{N}$, then $\mu(G_n) \downarrow \mu(G)$.

Note that $F_n \uparrow F$ means that $F_n \subseteq F_{n+1}$, $\forall n \in \mathbb{N}$, and $\bigcup_{n\in\mathbb{N}} F_n = F$.

Proof. To be done... □

Remark 1.30. The assumption in the second part of the previous theorem is necessary. We can take $H_n := (n, \infty)$, $n \in \mathbb{N}$ as a counterexample otherwise. Indeed, $H_n \downarrow \varnothing$, but $\mu(H_n) = \infty$, $\forall n \in \mathbb{N}$, where we take μ as the Lebesgue measure on $\mathbb{R}$.

1.2 Probability Spaces

We can think of the probability space $(\Omega, \mathscr{F}, \mathbb{P})$ as being a model for an experiment whose outcome is subject to chance. Ω is the set of all possible outcomes of the experiment; $\mathscr{F}$ is the set of observable outcomes (or events); and $\forall A \in \mathscr{F}$, $\mathbb{P}(A)$ is the probability of the event A, *i.e.*, the probability that the outcome $\omega \in \Omega$ of the experiment falls in the event A.

Example 1.31. (1) Let $(S, \mathscr{E}, \mu)$ be a measure space, and let $A \subseteq S$. Take $\mathscr{F} = \{A \cap C;\ C \in \mathscr{E}\}$. We shall be able to see that $\mathscr{F}$ is a σ-algebra on A, $\mu|_{\mathscr{F}}$ is a measure on $(A, \mathscr{F})$, and if $\mu(A) < \infty$, $\mathbb{P}(C) = \mu(C)/\mu(A)$ is a probability measure.

- $\varnothing \in \mathscr{F}$ because $\varnothing \in \mathscr{E}$. Take arbitrary $A \cap C \in \mathscr{F}$, then $(A \cap C)^C = A^C \cup C^C$. However, we are checking whether $\mathscr{F}$ is a σ-algebra on A, so this is just $C^C = A \cap C^C \in \mathscr{F}$ because $C^C \in \mathscr{E}$. Now take arbitrary $A \cap C_n \in \mathscr{F}$, then $\bigcup_{n\in\mathbb{N}}(A \cap C_n) = A \cap \left(\bigcup_{n\in\mathbb{N}} C_n\right) \in \mathscr{F}$ because $\bigcup_{n\in\mathbb{N}} C_n \in \mathscr{E}$. Therefore we have shown that $\mathscr{F}$ is a σ-algebra on $\mathscr{A}$.
- The properties of $\mu|_{\mathscr{F}}$ follow from that of μ.
- Clearly $\mathbb{P}(A) = 1$ and $\mathbb{P}(C) = \mu(C)/\mu(A) \leq \mu(C')/\mu(A) = \mathbb{P}(C')$ if $C \subseteq C'$. Therefore, it suffices to show that $\mathbb{P}$ is countably additive. Taking disjoint $C_n \in \mathscr{E}$, we have that

$$\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} C_n\right) = \frac{\mu\left(\bigcup_{n\in\mathbb{N}} C_n\right)}{\mu(A)} = \frac{\sum_{n\in\mathbb{N}} \mu(C_n)}{\mu(A)} = \sum_{n\in\mathbb{N}} \frac{\mu(C_n)}{\mu(A)} = \sum_{n\in\mathbb{N}} \mathbb{P}(C_n). \tag{29}$$

(2) On the measurable space $(S, \mathscr{P}(S))$ where $\mathscr{P}(S)$ is the power set of S, the **Dirac delta measure** or **unit mass** at $x \in S$ is defined as

$$\delta_x(A) := \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases} \tag{30}$$

9/6 Lecture

Example 1.32. Consider an infinite sequence of coin tosses. In this case, we may choose $\Omega = \{H, T\}^{\mathbb{N}}$. A sample point $\omega \in \Omega$ is a sequence $\omega = (\omega_n)_{n\geq 1}$ where $\omega_n = 0, 1$ (0 stands for H and 1 stands for T). Note that Ω is a very large space, and any useful probability measure can only be defined on a smaller σ-algebra than that of all subsets of Ω. Now let $A_l = \{\omega_l = 0\}$ be the event that 0 is tossed at time l, $\forall l \geq 1$. We further denote $A_l^1 = A_l$ and $A_l^0 = A_l^C$, and let $A_I^\epsilon := \bigcap_{k\in I} A_k^\epsilon$, $\epsilon \in \{0, 1\}$. For any $\epsilon \in \{0, 1\}^m$, let $A_\epsilon := A_1^{\epsilon_1} \cap \cdots \cap A_m^{\epsilon_m}$. Now for all $k \in \mathbb{N}$, $j \in \mathbb{N}^k$, and $x \in \{H, T\}^k$, let

$$E_{k,j,x} = \{\omega;\ \omega_{j_1} = x_1, \cdots, \omega_{j_k} = x_k\}, \tag{31}$$

which is the event that x_1 is tossed at time j_1, etc., and x_k is tossed at time j_k. Let

$$\mathscr{C} = \left\{E_{k,j,x};\ k \in \mathbb{N},\ j \in \mathbb{N}^k,\ x \in \{H, T\}^k\right\}, \tag{32}$$

where $\mathscr{C}$ is sometimes called the **set of cylinders**.

- $E_{k,j,x}$ can be written as an intersection of subsets of the type A_I^ϵ. Indeed, we can take $I_1 = \{j_i;\ x_i = H\}$ and $I_2 = \{j_i;\ x_i = T\}$, then $E_{k,j,x} = A_{I_1}^0 \cap A_{I_2}^1$.
- $\mathscr{C}$ is a π-system. Indeed, $\varnothing \in \mathscr{C}$ if we take $k = 0$. Moreover, we can see that

$$E_{k,j,x} \cap E_{k',j',x'} = A_{I_1}^0 \cap A_{I_2}^1 \cap A_{I_1'}^0 \cap A_{I_2'}^1 = A_{I_1 \cup I_1'}^0 \cap A_{I_2 \cup I_2'}^1 \in \mathscr{C}. \tag{33}$$

Let $\mathscr{E}$ be the smallest algebra containing $\mathscr{C}$, defined as the intersection of all algebras containing $\mathscr{C}$.

- $A \in \mathscr{E}$ if and only if there exists $m \geq 1$ and $K \subseteq \{0,1\}^m$ finite, such that $A = \bigcup_{\epsilon \in K} A_\epsilon$.

  $\Longleftarrow$ This is trivial. Note that we are dealing with algebras instead of σ-algebras, so we can extend to finite unions but not countable unions. Now each A_ϵ is a finite intersection of elements in $\mathscr{E}$, so $A_\epsilon \in \mathscr{E}$. Therefore, any finite union $A = \bigcup_{\epsilon \in K} A_\epsilon$ should clearly be still in $\mathscr{E}$.

  $\Longrightarrow$ Let

$$\mathscr{G} = \left\{ A;\ A = \bigcup_{\epsilon \in K} A_\epsilon \text{ for some } K \subseteq \{0,1\}^m \text{ finite, } m \geq 1 \right\}. \tag{34}$$

  Clearly $\varnothing \in \mathscr{G}$. Take an arbitrary $A = \bigcup_{\epsilon \in K} A_\epsilon \in \mathscr{G}$, then we have that

$$A^C = \bigcup_{\epsilon \in K^C} A_\epsilon \in \mathscr{G}. \tag{35}$$

  Note that the complement of K is taken for a fixed m. This is because for a fixed m, all A_ϵ's are disjoint, and their union is the entire space. Now take $A_1, A_2 \in \mathscr{G}$, then

$$A_1 \cup A_2 = \bigcup_{\epsilon \in K_1 \cup K_2} A_\epsilon \in \mathscr{G}, \tag{36}$$

  because K_1 and K_2 are both finite and thus $K \subseteq \{0,1\}^m$ is also finite. Therefore, we can conclude that $\mathscr{G}$ is an algebra. Note that $\mathscr{C} \subseteq \mathscr{G}$. Each $E_{k,j,x}$ is a fixed sequence of k coin tosses, and each A_ϵ represents a pattern for a subsequence of coin tosses. Their finite union is then multiple patterns for a subsequence of coin tosses. Clearly, each fixed sequence $E_{k,j,x}$ can find a suitable union to fall in. Now $\mathscr{G}$ is an algebra containing $\mathscr{C}$, while $\mathscr{E}$ is the smallest algebra containing $\mathscr{C}$. This means that $\mathscr{E} \subseteq \mathscr{G}$, and the proof is complete.

- $\mathbb{P}(E_{k,j,x}) := 2^{-k}$ (independent fair coin tosses) uniquely defines a probability measure $\mathbb{P}$ on $(\Omega, \mathscr{F})$, where $\mathscr{F} = \sigma(\mathscr{C})$ is the σ-algebra generated by $\mathscr{C}$. Uniqueness follows from Lemma 1.17, because $\mathscr{C}$ is a π-system as is previously shown. Now we check the existence. As we have previously shown, for each $A \in \mathscr{E}$, there exists $m \geq 1$ and $K \subseteq \{0,1\}^m$ finite, such that $A = \bigcup_{\epsilon \in K} A_\epsilon$ and $\mathbb{P}(A) = |K| 2^{-m}$. To see that this is self-consistent, suppose that $n \geq m$ and $A = \bigcup_{\epsilon' \in K \times \{0,1\}^{n-m}} A_{\epsilon'}$ is an alternative expression for A. Then we can check that

$$\mathbb{P}(A) = \frac{|K \times \{0,1\}^{n-m}|}{2^n} = \frac{|K| 2^{n-m}}{2^n} = \frac{|K|}{2^m}, \tag{37}$$

  so $\mathbb{P}$ is well-defined. In addition, $\mathbb{P}$ is countably additive since any countable union of $A_n \in \mathscr{E}$ can be written as $\bigcup_{\epsilon \in K} A_\epsilon$ with K being finite, so $\mathbb{P}(\bigcup_n A_n) < \infty$. By Carathéodory's extension theorem (Theorem 1.19), $\mathbb{P}$ is well-defined on $\mathscr{F} = \sigma(\mathscr{C})$.

Let $T : \{0,1\}^{\mathbb{N}} \to [0,1]$, such that $\omega \mapsto 0.\omega_1\omega_2\cdots\omega_n\cdots = \sum_{n=1}^\infty \omega_n 2^{-n}$.

- Let $\mathscr{B}([0,1])$ be the Borel σ-algebra on $[0,1]$, then $T^{-1}(\mathscr{B}([0,1])) \subseteq \mathscr{F}$. First we note that T is not bijective. For instance, $(1,0,0,\cdots)$ and $(0,1,1,\cdots)$ are both mapped to $1/2$, indicating that T is not injective and thus not bijective. We split the interval $[0,1]$. For instance, $T^{-1}([0,1/2)) = \{\omega_1 = 0\} \setminus \{\omega_2 = \omega_3 = \cdots = 1\}$ (the excluded part would give $1/2$ because the sequence is infinite). As a result, we observe that

$$\begin{aligned} &T^{-1}\left(\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right)\right) \\ &\quad = \underbrace{E_{n,\{1,\cdots,n\},x}}_{\text{start from } k2^{-n}} \setminus \{\omega_{n+1} = \omega_{n+2} = \cdots = 1\} \cup \{\omega_l = y_l \text{ for } 1 \leq l \leq n \text{ and } \omega_{n+1} = \omega_{n+2} = \cdots = 1\} \in \mathscr{F}, \end{aligned} \tag{38}$$

where we set x and y such that

$$k = \sum_{i=1}^{n} 2^{n-i} x_i, \qquad k+1 = \sum_{i=1}^{n} 2^{n-i} y_i. \tag{39}$$

Take some time to check this result.

- Let μ be the measure defined on $\mathscr{B}([0,1])$ by $\mu(I) = \mathbb{P}(T^{-1}(I))$, $I \in \mathscr{B}([0,1])$. Then we shall be able to see that μ is the Borel measure on $[0,1]$. To do this, we want to show that any open subset of $[0,1]$ can be expressed as a countable union of $[k2^{-n}, (k+1)2^{-n})$. Indeed, for any interval (a,b), we can write it as

$$(a,b) = \bigcup_{a < \frac{k}{2^n} < \frac{k+1}{2^n} \leq b} \left[\frac{k}{2^n}, \frac{k+1}{2^n}\right). \tag{40}$$

  Note that there always exists such $k \in \mathbb{Z}$. Indeed, let $a < k2^{-n} \leq x < (k+1)2^{-n} \leq b$, then there exists some integer k as long as $x - a > 2^{-n}$ and $b - x > 2^{-n}$. Need to apply uniqueness theorem on the π-system of these $[k2^{-n}, (k+1)2^{-n})$ intervals.

1.2.1 Independence

Definition 1.33. Let I be a countable set. We say that the events $\{A_i \in \mathscr{F};\ i \in I\}$ are **independent** if, for all finite subsets $J \subseteq I$, we have that

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i). \tag{41}$$

We say that the σ-algebras $\{A_i \subseteq \mathscr{F};\ i \in I\}$ are **independent** if $\{A_i \in \mathscr{F};\ i \in I\}$ are independent whenever $A_i \in \mathscr{A}_i$ for all $i \in I$.

Example 1.34. Let $\Omega := [0,1]^2$, $\mathscr{F} = \mathscr{B}([0,1]^2)$, and μ be the Borel measure on $[0,1]$. Let $\mathbb{P} := \mathscr{L}([0,1]^2)$ be the Lebesgue measure on $\mathbb{R}^2$, *i.e.*, the unique measure $\mathbb{P}$ such that $\mathbb{P}(A_1 \times A_2) = \mu(A_1)\mu(A_2)$, $\forall A_1, A_2 \in \mathscr{B}([0,1])$. Now let $\mathscr{E}_1 := \{A \times [0,1];\ A \in \mathscr{B}([0,1])\}$ and $\mathscr{E}_2 := \{[0,1] \times B;\ B \in \mathscr{B}([0,1])\}$, then $\mathscr{E}_1$ and $\mathscr{E}_2$ are independent on $(\Omega, \mathscr{F}, \mathbb{P})$. Indeed, take arbitrary $G = A \times [0,1] \in \mathscr{E}_1$ and $H = [0,1] \times B \in \mathscr{E}_2$, we can see that

$$\mathbb{P}(G \cap H) = \mathbb{P}(A \times B) = \mu(A)\mu(B) = \mathbb{P}(A \times [0,1])\mathbb{P}([0,1] \times B) = \mathbb{P}(G)\mathbb{P}(H). \tag{42}$$

Example 1.35. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of events. Show that A_n, $n \in \mathbb{N}$ are independent if and only if the σ-algebras they generate, *i.e.*, $\sigma(A_n) = \left\{\varnothing, A_n, A_n^C, \Omega\right\}$, are independent.

$\Longleftarrow$ This is immediate by definition.

$\Longrightarrow$ It suffices to show that for all finite $J, K \subset \mathbb{N}$ disjoint, we have that

$$\mathbb{P}\left(\left(\bigcap_{n \in J} A_n\right) \cap \left(\bigcap_{n \in K} A_n^C\right)\right) = \prod_{n \in J} \mathbb{P}(A_n) \prod_{n \in K} \mathbb{P}(A_n^C). \tag{43}$$

We prove by mathematical induction on $|K|$. If $|K| = 0$, then this holds trivially by independence of $\{A_n\}_{n \in \mathbb{N}}$. Now assume that the above holds for all $K \subset \mathbb{N}$ with $|K| = N$, then for $|K| = N+1$, we can write $K = K_0 \cup \{j\}$.

Using the inductive hypothesis, we can write that

$$\begin{aligned}
\mathbb{P}\left(\left(\bigcap_{n\in J} A_n\right)\cap\left(\bigcap_{n\in K} A_n^C\right)\right) &= \mathbb{P}\left(\left(\bigcap_{n\in J} A_n\right)\cap\left(\bigcap_{n\in K_0} A_n^C\right)\cap A_j^C\right) \\
&= \mathbb{P}\left(\left(\bigcap_{n\in J} A_n\right)\cap\left(\bigcap_{n\in K_0} A_n^C\right)\right) - \mathbb{P}\left(\left(\bigcap_{n\in J} A_n\right)\cap\left(\bigcap_{n\in K_0} A_n^C\right)\cap A_j\right) \\
&= \mathbb{P}\left(\left(\bigcap_{n\in J} A_n\right)\cap\left(\bigcap_{n\in K_0} A_n^C\right)\right) - \mathbb{P}\left(\left(\bigcap_{n\in J\cup\{j\}} A_n\right)\cap\left(\bigcap_{n\in K_0} A_n^C\right)\right) \\
&= \underbrace{\prod_{n\in J}\mathbb{P}(A_n)\prod_{n\in K_0}\mathbb{P}(A_n^C)}_{\text{inductive hypothesis}} - \underbrace{\prod_{n\in J\cup\{j\}}\mathbb{P}(A_n)\prod_{n\in K_0}\mathbb{P}(A_n^C)}_{\text{inductive hypothesis}} \\
&= \left(\prod_{n\in J}\mathbb{P}(A_n)\prod_{n\in K_0}\mathbb{P}(A_n^C)\right)(1-\mathbb{P}(A_j)) = \left(\prod_{n\in J}\mathbb{P}(A_n)\prod_{n\in K_0}\mathbb{P}(A_n^C)\right)\mathbb{P}(A_j^C) \\
&= \prod_{n\in J}\mathbb{P}(A_n)\prod_{n\in K_0\cup\{j\}}\mathbb{P}(A_n^C) = \prod_{n\in J}\mathbb{P}(A_n)\prod_{n\in K}\mathbb{P}(A_n^C). \qquad (44)
\end{aligned}$$

Therefore, we can conclude the proof by induction.

Theorem 1.36. Suppose that $\mathscr{G}$ and $\mathscr{H}$ are sub-σ-algebras of $\mathscr{F}$, and that $\mathscr{G}_0$ and $\mathscr{H}_0$ are π-systems with $\sigma(\mathscr{G}_0)=\mathscr{G}$ and $\sigma(\mathscr{H}_0)=\mathscr{H}$. Then $\mathscr{G}$ and $\mathscr{H}$ are independent if and only if $\mathscr{G}_0$ and $\mathscr{H}_0$ are independent.

Proof. $\implies$ This is trivial since $\mathscr{G}_0\subseteq\mathscr{G}$ and $\mathscr{H}_0\subseteq\mathscr{H}$.

$\impliedby$ Fix $G\in\mathscr{G}_0$, then take $H\mapsto\mathbb{P}(G\cap H)$ and $H\mapsto\mathbb{P}(G)\mathbb{P}(H)$. They are clearly equal since $\mathscr{G}_0$ and $\mathscr{H}_0$ are independent, and they have the same total mass since $\mathbb{P}(G\cap\Omega)=\mathbb{P}(G)\mathbb{P}(\Omega)=\mathbb{P}(G)$. By the uniqueness (Lemma 1.17), they should be equal on $\mathscr{H}$. Now fix $H\in\mathscr{H}_0$ and analogously repeat the previous arguments. We can conclude that $\mathbb{P}(G\cap H)=\mathbb{P}(G)\mathbb{P}(H)$ for any $G\in\mathscr{G}$ and $H\in\mathscr{H}$, so $\mathscr{G}$ and $\mathscr{H}$ are independent. □

1.2.2 The Fatou and Borel-Cantelli Lemmas

For the rest of this section, let $(E,\mathscr{E},\mu)$ be a measure space and let $\{A_n\}_{n\in\mathbb{N}}$ be a sequence in $\mathscr{E}$.

Definition 1.37. We define the **limit superior** and **limit inferior** of the sequence $\{A_n\}_{n\in\mathbb{N}}$ respectively as

$$\limsup_{n\to\infty} A_n := \bigcap_{n\in\mathbb{N}}\bigcup_{m\geq n} A_m = \{A_n \text{ occurs infinitely often } (i.o.)\}, \qquad (45)$$

$$\liminf_{n\to\infty} A_n := \bigcup_{n\in\mathbb{N}}\bigcap_{m\geq n} A_m = \{A_n \text{ occurs at large times}\}. \qquad (46)$$

Remark 1.38. If $(E,\mathscr{E},\mu)$ is a probability space, then we can interpret $\limsup_{n\to\infty} A_n$ as being the event on which A_n occurs for infinitely many n, and $\liminf_{n\to\infty} A_n$ as being the event on which there exists some (random) N such that A_n occurs for all $n\geq N$. Note that we always have $\liminf_{n\to\infty} A_n\subset\limsup_{n\to\infty} A_n$.

Example 1.39. Check that

$$\mathbb{1}_{\limsup A_n} = \limsup \mathbb{1}_{A_n}, \qquad \mathbb{1}_{\liminf A_n} = \liminf \mathbb{1}_{A_n}. \qquad (47)$$

9/8 Recitation

Example 1.40. Let $\Omega=\mathbb{R}$ and $\mathscr{F}$ be a set of all $A\in\Omega$ such that either A or A^C is countable. Define

$$\mathbb{P}(A) = \begin{cases} 0, & \text{if } A \text{ is countable,} \\ 1, & \text{if } A^C \text{ is countable.} \end{cases} \qquad (48)$$

Show that $(\Omega,\mathscr{F},\mathbb{P})$ forms a probability space.

Proof. We first need to show that $\mathscr{F}$ is a σ-algebra on Ω. Clearly, $\varnothing \in \Omega$ since $\varnothing$ is countable. Take arbitrary $A, A_n \in \mathscr{F}$, then $A^C \in \mathscr{F}$ because if A is countable then A^C is countable, and otherwise $(A^C)^C = A$ is countable. Moreover, $\bigcup_{n\in\mathbb{N}} A_n \in \mathscr{F}$ because if all A_n's are countable then their countable union is still countable, and otherwise we can see that $\left(\bigcup_{n\in\mathbb{N}} A_n\right)^C = \bigcap_{n\in\mathbb{N}}(A_n)^C$ is countable since some $(A_n)^C$ would then be countable. Up till now we have shown that $\mathscr{F}$ is a σ-algebra on Ω. Now we check that $\mathbb{P}$ is a countably additive measure. Take arbitrary $A_n \in \mathscr{F}$ disjoint, $n \in \mathbb{N}$, then there are two cases.

- If all A_n's are countable, then $\bigcup_{n\in\mathbb{N}} A_n$ is countable, so $0 = \mathbb{P}\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} \mathbb{P}(A_n) = 0$.
- If there exists A_{n_0} uncountable, then $\bigcup_{n\in\mathbb{N}} A_n$ is uncountable. Since the other A_n's are disjoint with A_{n_0}, each of them belongs to $A_{n_0}^C$, but note that $A_{n_0}^C$ is countable. Therefore, all other A_n's must be countable, so $1 = \mathbb{P}\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\neq n_0} \mathbb{P}(A_n) + \mathbb{P}(A_{n_0}) = 1$.

This means that $\mathbb{P}$ is countably additive. Furthermore, $\Omega^C = \varnothing$ is countable, so that $\mathbb{P}(\Omega) = 1$, meaning that $\mathbb{P}$ is a probability measure, so the proof is complete. □

Example 1.41. Let $\Omega = \{1, 2, 3, 4\}$, $\mathscr{F} = \mathscr{P}(\Omega)$, and $\mathbb{P}$ the probability measure such that

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}). \tag{49}$$

(1) Show that the two classes of events $\mathscr{C}_1 = \{\{1, 2\}\}$ and $\mathscr{C}_2 = \{\{2, 3\}, \{2, 4\}\}$ are independent by $\sigma(\mathscr{C}_1)$ and $\sigma(\mathscr{C}_2)$ are not.

(2) We have learned a theorem that stablish an extra condition in which we can conclude $\sigma(\mathscr{C}_1)$ and $\sigma(\mathscr{C}_2)$ are independent. Identify the condition that is missing in this example $\mathscr{C}_1$ and $\mathscr{C}_2$.

Proof. Computing the probabilities of the intersections, we have that

$$\mathbb{P}(\{1, 2\} \cap \{2, 3\}) = \mathbb{P}(\{2\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(\{1, 2\}) \cdot \mathbb{P}(\{2, 3\}), \tag{50}$$

$$\mathbb{P}(\{1, 2\} \cap \{2, 4\}) = \mathbb{P}(\{2\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(\{1, 2\}) \cdot \mathbb{P}(\{2, 4\}). \tag{51}$$

Therefore, $\mathscr{C}_1$ and $\mathscr{C}_2$ are independent. Now note that $\sigma(\mathscr{C}_1) = \{\varnothing, \{1, 2\}, \{3, 4\}, \Omega\}$, and $\sigma(\mathscr{C}_2) = \mathscr{P}(\Omega)$. Then, we can see that

$$\mathbb{P}(\underbrace{\{1, 2\}}_{\in\sigma(\mathscr{C}_1)} \cap \underbrace{\{1, 2\}}_{\in\sigma(\mathscr{C}_2)}) = \mathbb{P}(\{1, 2\}) = \frac{1}{2} \neq \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(\underbrace{\{1, 2\}}_{\in\sigma(\mathscr{C}_1)}) \cdot \mathbb{P}(\underbrace{\{1, 2\}}_{\in\sigma(\mathscr{C}_2)}), \tag{52}$$

which means that $\sigma(\mathscr{C}_1)$ and $\sigma(\mathscr{C}_2)$ are not independent. To answer the second part of this question, this is because $\mathscr{C}_2$ is not a π-system (clearly $\{2, 3\} \cap \{2, 4\} = \{2\} \notin \mathscr{C}_2$). □

Example 1.42. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and X be a measurable function. Show that μ defined in $(\mathbb{R}, \mathscr{B})$ given by $\mu(B) = \mathbb{P}(X^{-1}(B))$ is a probability measure.

Proof. To see that μ is a probability measure, we need to check that it is countably additive. Take arbitrary $B_n \in \mathscr{B}$ disjoint, $n \in \mathbb{N}$, then we have that

$$\mu\left(\bigcup_{n\in\mathbb{N}} B_n\right) = \mathbb{P}\left(X^{-1}\left(\bigcup_{n\in\mathbb{N}} B_n\right)\right) = \mathbb{P}\left(\bigcup_{n\in\mathbb{N}} X^{-1}(B_n)\right) = \sum_{n\in\mathbb{N}} \mathbb{P}(X^{-1}(B_n)) = \sum_{n\in\mathbb{N}} \mu(B_n), \tag{53}$$

so μ is indeed countably additive. Clearly $\mu(\mathscr{B}) = \mathbb{P}(X^{-1}(\mathscr{B})) = \mathbb{P}(\Omega) = 1$, so μ is a probability measure and the proof is complete. □

Remark 1.43. In the above proof, we have used that $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$. This is indeed true.

$\subseteq$ Take an arbitrary $x \in f^{-1}(A \cup B)$, then $f(x) \in A \cup B$. This means that either $f(x) \in A$ or $f(x) \in B$. Then either $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, implying that $x \in f^{-1}(A) \cup f^{-1}(B)$.

$\supseteq$ Take an arbitrary $x \in f^{-1}(A) \cup f^{-1}(B)$, then either $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$. This means that either $f(x) \in A$ or $f(x) \in B$, implying that $f(x) \in A \cup B$. Thus $x \in f^{-1}(A \cup B)$.

Example 1.44. Prove that, if $\{A_n\}_{n\geq 1}$ are independent events, then

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \prod_{n=1}^{\infty} \mu(A_n). \tag{54}$$

Proof. Let $B_k = \bigcap_{n=1}^{k} A_n$, then by independence, we have that

$$\mathbb{P}(B_k) = \mathbb{P}\left(\bigcap_{n=1}^{k} A_n\right) = \prod_{n=1}^{k} \mathbb{P}(A_n). \tag{55}$$

Note that B_n is a decreasing event, such that $B_n \downarrow \bigcap_{n=1}^{\infty} A_n$. By monotone convergence, we thus have that

$$\lim_{k\to\infty} \mathbb{P}(B_k) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right). \tag{56}$$

Therefore, we can conclude that

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{k\to\infty} \prod_{n=1}^{k} \mathbb{P}(A_n) = \prod_{n=1}^{\infty} \mathbb{P}(A_n), \tag{57}$$

and the proof is complete. □

Example 1.45. Prove that, if $\{A_n\}_{n\geq 1}$ are independent events, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = 1 - \prod_{n=1}^{\infty} (1 - \mu(A_n)). \tag{58}$$

Proof. We know that if $\{A_n\}_{n\geq 1}$ are independent, then $\left\{A_n^C\right\}_{n\geq 1}$ are independent as well. Therefore, using the conclusion of Example 1.44, we can deduce that

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = 1 - \mu\left(\bigcap_{n=1}^{\infty} A_n^C\right) = 1 - \prod_{n=1}^{\infty} \mu(A_n^C) = 1 - \prod_{n=1}^{\infty} (1 - \mu(A_n)), \tag{59}$$

so the proof is complete. □

Example 1.46. Prove that

$$\limsup_{n\to\infty} A_n^C = \left(\liminf_{n\to\infty} A_n\right)^C, \qquad \limsup_{n\to\infty}\left(\liminf_{k\to\infty}(A_n \cap A_k^C)\right) = \varnothing. \tag{60}$$

Proof. By definition, we have that

$$\left(\liminf_{n\to\infty} A_n\right)^C = \left(\bigcup_{n\geq 1}\bigcap_{m\geq n} A_m\right)^C = \bigcap_{n\geq 1}\bigcup_{m\geq n} A_m^C = \limsup_{n\to\infty} A_n^C, \tag{61}$$

$$\begin{aligned}\limsup_{n\to\infty}\left(\liminf_{k\to\infty}(A_n \cap A_k^C)\right) &= \limsup_{n\to\infty}\left(\bigcup_{k\geq 1}\bigcap_{p\geq k}(A_n \cap A_p^C)\right) = \limsup_{n\to\infty}\left(A_n \cap \left(\bigcup_{k\geq 1}\bigcap_{p\geq k} A_p^C\right)\right) \\ &= \limsup_{n\to\infty}\left(A_n \cap \liminf_{k\to\infty} A_k^C\right) = \left(\limsup_{n\to\infty} A_n\right) \cap \left(\liminf_{k\to\infty} A_k^C\right) = \varnothing,\end{aligned} \tag{62}$$

so the proof is complete. □

9/11 Lecture

Lemma 1.47 (Reverse Fatou lemma for sets)**.** Assume that μ is finite, then $\mu(\limsup A_n) \leq \limsup \mu(A_n)$.

Proof. Let $G_m := \bigcup_{n\geq m} A_n$, then $G_m \downarrow \limsup A_n$ and therefore $\mu(G_m) \downarrow \mu(\limsup A_n)$ by the monotone convergence theorem (Theorem 1.29). Now note that $\mu(G_m) \geq \sup_{n\geq m} \mu(A_n)$ because G_m is larger than any of them. Therefore by taking $m \to \infty$ on both sides of the inequality, we can conclude that

$$\mu(\limsup_{n\to\infty} A_n) \geq \lim_{m\to\infty} \sup_{n\geq m} \mu(A_n) = \limsup_{n\to\infty} \mu(A_n), \tag{63}$$

and the proof is complete. □

Lemma 1.48 (The first Borel-Cantelli lemma)**.** Assume $\sum_{n\in\mathbb{N}} \mu(A_n) < \infty$, then $\mu(\limsup A_n) = \mu(\{A_n \ i.o.\}) = 0$.

Proof. Let $G_m := \bigcup_{n\geq m} A_n$, then we have that $\mu(G_m) \leq \sum_{n\geq m} \mu(A_n) < \infty$. Note that $G_m \downarrow \limsup A_n$ and therefore $\mu(G_m) \downarrow \mu(\limsup A_n)$ by the monotone convergence theorem (Theorem 1.29), bring $m \to \infty$ so that

$$\mu(\limsup_{n\to\infty} A_n) \leq \lim_{m\to\infty} \sum_{n\geq m} \mu(A_n) = 0, \tag{64}$$

and the proof is complete. □

Lemma 1.49 (The second Borel-Cantelli lemma)**.** Assume that $\{A_n\}_{n\in\mathbb{N}}$ are independent events. If $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(\limsup A_n) = \mathbb{P}(\{A_n \ i.o.\}) = 1$

Proof. By independence, for all m we have that

$$\mathbb{P}\left(\bigcap_{n\geq m} A_n^C\right) = \prod_{n\geq m} (1 - \mathbb{P}(A_n)) \leq \prod_{n\geq m} \exp\left(-\mathbb{P}(A_n)\right) = \exp\left(-\sum_{n\geq m} \mathbb{P}(A_n)\right) = 0. \tag{65}$$

Therefore, we can deduce that

$$1 \geq \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = \mathbb{P}\left(\bigcap_{m\in\mathbb{N}} \bigcup_{n\geq m} A_n\right) = 1 - \mathbb{P}\left(\bigcup_{m\in\mathbb{N}} \bigcap_{n\geq m} A_n^C\right) \geq 1 - \sum_{m\in\mathbb{N}} \mathbb{P}\left(\bigcap_{n\geq m} A_n^C\right) = 1, \tag{66}$$

which completes the proof. □